

10-User E2E Live Runtime and World Generation Report

Final English report based on the repaired July 7, 2026 stress-test results for Clockwork Rain Conservatory, covering world generation, art readiness, REST live runtime, WebSocket movement runtime, and the root-cause fix.

Final conclusion: world generation, FLUX art, and publish readiness are stable. The 10-user REST and 10-user WebSocket live runtime paths have returned to low-latency behavior. The earlier 16-90 second latency was not caused by the live store, AI model changes, or artificial sleeps; it was caused by a route guard that triggered full catalog/latest-revision scanning on every live request.

Date

2026-07-07 UTC

Target world

Clockwork Rain Conservatory

Draft / Revision

creator_20260706_053251_1ad0b086 / r001

Validation access code

70c1c22446644136

Final REST artifact

export_artifact/live_load_tests/20260707_10user_rest_clockwork_rain_live_route_fix/

Final WS artifact

export_artifact/live_load_tests/20260707_10user_ws_clockwork_rain_final/

1. Executive Summary

World Draft Ready Draft manifest is publish-ready.	Art Pipeline Ready FLUX, pixel-read, and map QA passed.	10-user REST Pass Zero errors; stress signal false.	10-user WS Pass 10/10 connected; 898 deltas.
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The final test results confirm that Agora 2.0's world generation and live runtime architecture are intact. The initial timeout pattern came from route-layer validation, not from the live runtime itself. After the fix, REST live action p50 returned to 15ms, and WebSocket movement delta p50 returned to 28ms.

2. Root Cause

Faulty hot-path call chain: high-frequency live endpoint ->

_require_latest_pixel_world_access_code() -> _canonical_pixel_world_record() ->
_latest_pixel_world_records() -> scan/materialize public package catalog.

Observation	Evidence	Interpretation
Direct store profiling was fast	create_session ~70ms; heartbeat, state, and action roughly 1-2ms	The core live store was not the bottleneck.
Route validation was slow	_require_latest_pixel_world_access_code() cost roughly 7.4-8.0s per call	Every live request paid full catalog validation cost.

WebSocket handshake timed out	Before the fix, 10/10 WS handshakes exceeded timeout	The WS route used the same slow canonical-world guard before accepting.
No artificial delay matched the symptom	No business-path sleep explained the repeated 16-90s pattern	The regression was validation architecture, not pacing logic.

3. Fixes Applied

File	Change	Runtime effect
macro_ui/ routes/ pixel_api.py	Added <code>_require_pixel_live_access_code()</code> and switched live session, state, action, heartbeat, release, and file routes to it.	REST live endpoints no longer scan the full catalog.
macro_ui/ routes/ websocket.py	Switched WebSocket route validation to the lightweight live access-code check.	WebSocket handshakes can accept promptly.
macro_ui/ serve_macro_ui.py	Made uvicorn reload opt-in via <code>--reload</code> or <code>AGORA_MACRO_UI_RELOAD=1</code> , and narrowed reload directories when enabled.	Load tests avoid dev reloader overhead and package-tree watching.
scripts/ pixel_live_load_test.py	WebSocket <code>state_delta</code> is treated as transport success; movement rejection remains recorded as <code>accepted=false</code> .	The stress summary no longer misclassifies collision or rule rejection as backend failure.

4. Post-Fix REST 10-User Result

Transport: `rest-mixed`. Users created: **10/10**. Duration: 45 seconds. Backend stress signal: **false**.

Endpoint	Count	Error rate	p50 ms	p95 ms	p99 ms	Max ms
create_session	10	0.0	81.239	228.508	303.974	322.841

heartbeat	120	0.0	43.837	147.796	161.311	202.189
live_state	352	0.0	56.901	167.830	256.580	332.663
live_action	60	0.0	14.776	108.593	111.508	112.306

REST conclusion: state, action, and heartbeat endpoints all returned to the expected low-latency envelope. The earlier 90-second timeouts were resolved by removing full catalog validation from live paths.

5. Post-Fix WebSocket 10-User Result

Transport: ws-movement. Users created: **10/10**. Duration: 25 seconds. Backend stress signal: **false**.

Endpoint	Count	Error rate	p50 ms	p95 ms	p99 ms	Max ms
create_session	10	0.0	81.598	230.464	306.222	325.162
ws_connect	10	0.0	520.231	630.014	644.895	648.615
ws_move_delta	898	0.0	28.369	51.555	54.887	211.130
ws_error	0	0.0	n/a	n/a	n/a	n/a

WebSocket conclusion: handshake and realtime movement are stable at 10 concurrent users. The p50 movement delta is 28ms and the p95 movement delta is 52ms.

6. World Generation And Art Pipeline Status

Area	Status	Evidence
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Draft generation	publish_ready	Planner, rooms, items, roles, profiles, inventory, compiler report, and package were generated successfully.
Generated world shape	valid	8 rooms, 25 agents, 25 main characters, 8 ordinary routes, 1 cinematic route, and 3 gameplay loops.
Inventory and profile generation	recovered	One invalid JSON retry in profile/inventory stages recovered; fallback-only inventory was not accepted.
Art pipeline	publish_ready	FLUX world asset generation, live-ready feed, Firefox pixel launch, and map QA all passed.
Package runtime	valid	<code>pixel_read=true</code> , backend startup ok, <code>agent_count=25</code> , <code>live_ready_count=25</code> .

7. Before/After Summary

Path	Before fix	After fix
REST create_session	p50 about 4.5s	p50 81ms
REST live_state/action/heartbeat	single-user calls around 16-19s; 10-user run timed out	p50 15-57ms; zero errors
WebSocket connect	10/10 handshakes timed out or failed	10/10 connected; p50 520ms
WebSocket movement	No valid movement stream	898 deltas; p50 28ms
Active test-session cleanup	Failed HTTP release could leave active Load Tester sessions	Active Load Tester count verified as 0

8. Current Readiness

- **World creator:** ready for the publish flow.
- **Art:** ready; the current FLUX-based pipeline passed map QA and pixel-read validation.

- **REST live runtime:** ready for the 10-user API/interaction E2E harness.
- **WebSocket movement:** ready for the 10-user movement harness.
- **Not yet covered:** 10 simultaneous full Firefox canvas clients rendering Pixel UI.

9. Recommendations

Priority	Recommendation	Reason
P0	Keep full latest/public catalog validation off live hot paths.	This was the direct root cause of the latency regression.
P1	Run a 10-browser Firefox canvas E2E test next.	API and WebSocket latency are healthy again; browser rendering and asset hydration are the remaining realistic concurrency surface.
P1	Consider compact session-create and WebSocket hello payloads later.	The initial payload is still around 600KB. It no longer breaks latency, but it is worth optimizing.
P2	Add a local-mode cleanup fallback to the load harness.	If HTTP release times out, use <code>PixelLiveStore.release_session()</code> to avoid active test users.

10. Artifact Index

Artifact	Path
Markdown technical report	docs/technical_report_20260707_e2e10_world_generation.md
REST final summary	export_artifact/live_load_tests/20260707_10user_rest_clockwork_rain_live_route_fix/summary.json
WebSocket final summary	export_artifact/live_load_tests/20260707_10user_ws_clockwork_rain_final/summary.json
Draft manifest	output/world_creator_drafts/creator_20260706_053251_1ad0b086/

draft_manifest.json

Art status

output/world_creator_drafts/creator_20260706_053251_1ad0b086/revisions/
r001/art_status.json

Report generated for Agora 2.0 on 2026-07-07. This English PDF reflects the repaired route-layer validation architecture and the final clean 10-user REST/WebSocket stress summaries, not the earlier pre-fix timeout runs.